$= 2(0)\ \ln(0)\ \cos\theta$

$= (0)(\infty)\ \sin\theta$

$= 0$ Ans

Exercise 18.7

Tangent planes & normal vectors

Example 2

Find an equation for the tangent plane & parametric eq. for the normal line to the surface $z = x^2y$ at point $(2, 1, 4)$

Sol:-

$$z = x^2 y$$

$f(x,y,z) = z - x^2 y$

$\frac{\partial f}{\partial x} = fx = -2xy$, $fx|(2,1,4) = -4$

$\frac{\partial f}{\partial y} = fy = -x^2$, $fy|(2,1,4) = -4$

$\frac{\partial f}{\partial z} = fz = 1$, $fz|(2,1,4) = 1$

$\Delta f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}$

$= -4\hat{i} + -4j + k$

Eq. for tangent plane:

$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$

$-4(x-2) - 4(y-1) + 1(z-4) = 0$

$-4x + 8 - 4y + 4 + z - 4 = 0$

$-4x - 4y + z + 8 = 0$

Parametric eq. for normal line

$x = x_0 + at \Rightarrow x = 2 - 4t$

$y = y_0 + bt \Rightarrow y = 1 - 4t$

$z = z_0 + ct \Rightarrow z = 4 + t$

Example 3

Find parametric eqs of tangent line to the curve of intersection of paraboloid $z = x^2 + y^2$ ellipsoid $3x^2 + 2y^2 + z^2 = 9$ at point $(1,1,2)$

Sol:-

$f(x,y,z) = x^2 + y^2 - z$

$g(x,y,z) = 3x^2 + 2y^2 + z^2 - 9$

$f_x = \frac{\partial f}{\partial x} = 2x$, $f_x|(1,1,2) = 2$

$f_y = \frac{\partial f}{\partial y} = 2y$, $f_y|(1,1,2) = 2$

$f_z = \frac{\partial f}{\partial z} = -1$, $f_y|(1,1,2) = -1$.

$g_x = \frac{\partial g}{\partial x} = 6x$, $g_x|(1,1,2) = 6$

$g_y = \frac{\partial g}{\partial y} = 4y$, $g_y|(1,1,2) = 4$

$g_z = \frac{\partial g}{\partial z} = 2z$, $g_z|(1,1,2) = 4$

$\Delta f = f_x \hat{i} + f_y \hat{j} + f_z \hat{k}$
$= 2\hat{i} + 2\hat{j} - 1\hat{k}$

$\Delta g = g_x \hat{i} + g_y \hat{j} + g_z \hat{k}$
$= 6\hat{i} + 4\hat{j} + 4k$

$$\Delta f \times \Delta G = \begin{vmatrix} i & j & k \\ 2 & 2 & -1 \\ 6 & 4 & 4 \end{vmatrix}$$

$= i(8+4) - j(8+6) + k(8-12)$
$= 12\hat{i} - 14\hat{j} - 4\hat{k} \Rightarrow 6\hat{i} - 7\hat{j} - 2\hat{k}$

parametric eq
$x = x_0 + at = 1 + 12t$ or $1 + 6t$
$y = y_0 + bt = 1 - 14t$ or $1 - 7t$
$z = z_0 + ct = 2 - 4t$ or $1 - 2t$

Ex 13.7
Example 1, 2 and 3
Exercise: Q9, 10, Q29, 30

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Example 1:

Consider the ellipsoid $x^2 + 4y^2 + z^2 = 18$

(a) Find an eq. of the tangent plane to the ellipsoid at the point $(1, 2, 1)$

(b) Find parametric eq. of the line that is normal to the ellipsoid at the point $(1, 2, 1)$

(c) Find the acute angle ~~of~~ that the tangent plane at the point $(1, 2, 1)$ makes with xy-plane

Sol:-

(a) $f(x, y, z) = x^2 + 4y^2 + z^2 - 18$

$f_x = 2x \quad , \quad f_x|(1, 2, 1) = 2$

$f_y = 8y \quad , \quad f_y|(1, 2, 1) = 16$

$f_z = 2z \quad , \quad f_z|(1, 2, 1) = 2$

$$\Delta f = 2\hat{i} + 16\hat{j} + 2\hat{k}$$

Eq. of tangent

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

$$2(x - 1) + 16(y - 2) + 2(2 - 1) = 0$$

$$2x - 2 + 16y - 32 + 2z - 2 = 0$$

$$2x + 16y + 2z = 36$$

$$x + 8y + z = 18$$

(b) parametric eq.

$$x = x_0 + at = 1 + 2t$$

$$y = y_0 + bt = 2 + 16$$

$$z = z_0 + ct = z = 2t$$

(c) $\Delta f \cdot$ vector $\quad x = 0,\ y = 0,\ z = 1$

$(0, 0, 1)$

$$\cos\theta = \frac{|(2, 16, 2) \cdot (0, 0, 1)|}{\|2, 16, 2\| \, \|0, 0, 1\|}$$

$$\cos\theta = \frac{2}{(2\sqrt{66})(1)} = \frac{1}{\sqrt{66}}$$

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{66}}\right) = 83^\circ$$

Ex. 13.7 (tangent and parametric)

Q9) $z = xe^{-y}$; $P(1,0,1)$

Sol:-

$f(x,y,z) = z = xe^{-y} \Rightarrow xe^{-y} - z$

$f_x = e^{-y}$ $\quad f_x|(1,0,1) = 1$

$f_y = -xe^{-y}$ $\quad f_y|(1,0,1) = -1$

$f_z = -1$ $\quad f_z|(1,0,1) = -1$

$\Delta f = f_x\hat{i} + f_y\hat{j} + f_z\hat{k}$

$\Delta f = \hat{i} - \hat{j} + \hat{k}$

Eq. for tangent

$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$

$1(x-1) - 1(y-0) - 1(z-1) = 0$

$= x-1-y-z+1 = 0$

$= x-y-z = 0$

parametric eq

$x = x_0 + at = 1+t$

$y = y_0 + bt = 0 - t$

$z = z_0 + ct = 1-t$

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Q10) $z = \ln\sqrt{x^2+y^2}$; $P(-1,0,0)$

Sol:-

$f(x,y,z) = \ln\sqrt{x^2+y^2} - z$

$f_x = \dfrac{2x}{2\sqrt{x^2+y^2}}$, $f_x|(-1,0,0) = -1$

$f_y = \dfrac{2y}{2\sqrt{x^2+y^2}}$, $f_y|(-1,0,0) = 0$

$f_z = -1$, $f_z|(-1,0,0) = -1$

Eq of tangent $\nabla f = -\hat{i} - \hat{k}$

$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$

$-x + 0 + 0 - 1(z-0) = 0$

$-z = 0$

parametric eq.

$x = x_0 + at = -1-t$

$y = y_0 + bt = 0$

$z = z_0 + ct = -t$

Q29) Find the parametric eq. for the tangent line to the curve of intersection of the paraboloid $z = x^2 + y^2$ and the ellipsoid $x^2 + 4y^2 + z^2 = 9$ at the point $(1, -1, 2)$

Sol:-

$f(x, y, z) = x^2 + y^2 - z$

$g(x, y, z) = x^2 + 4y^2 + z^2 - 9$

$f_x = 2x \quad , \quad f_x|(1,-1,2) = 2$

$f_y = 2y \quad , \quad f_y|(1,-1,2) = -2$

$f_z = -1 \quad , \quad f_z(1,-1,2) = -1$

$g_x = 2x \quad , \quad g_x|(1,-1,2) = 2$

$g_y = 8y \quad , \quad g_y|(1,-1,2) = -8$

$g_z = 2z \quad , \quad g_z|(1,-1,2) = 4$

$\Delta f = f_x \hat{i} + f_y \hat{j} + f_z \hat{k}$

$= 2\hat{i} - 2\hat{j} - \hat{k}$

$\Delta g = g_x \hat{i} + g_y \hat{j} + g_z \hat{k}$

$= 2\hat{i} - 8\hat{j} + 4\hat{k}$

$$\Delta f \times \Delta g = \begin{vmatrix} i & j & k \\ 2 & -2 & -1 \\ 2 & -8 & 4 \end{vmatrix}$$

$$= \hat{i}(-8 - 8) + \hat{j}(-8 - 2) + \hat{k}(-16+4)$$

$$= -16\hat{i} - 10\hat{j} - 12\hat{k}$$

30) Find parametric equations for the tangent line to the curve of intersection of the ~~cylinders $x^2+z^2=2$ and $y^2+z^2=25$ at the point (3,3,4)~~ ~~Sol:-~~ ~~x^2~~ ~~$z=$~~ cone $z = \sqrt{x^2+y^2}$ and the plane $x+2y+2z=20$ at the point $(4,3,5)$.

Sol:-

$$f(x,y,z) = (x^2+y^2)^{1/2} - z$$

$$g(x,y,z) = x + 2y + 2z - 20$$

$$f_x = \frac{2x}{2\sqrt{x^2+y^2}} \Rightarrow f_x = \frac{x}{\sqrt{x^2+y^2}}$$

$$f_y = \frac{2y}{2\sqrt{x^2+y^2}} \Rightarrow f_y = \frac{y}{\sqrt{x^2+y^2}}$$

$$f_z = -1$$

~~f_x~~

$gx = 1$
$gy = 2$
$gz = 2$

$$f_x|_{(4,3,5)} = \frac{4}{\sqrt{(4^2)+(3)^2}} \Rightarrow \frac{4}{5}$$

$$f_y|_{(4,3,5)} = \frac{3}{5}$$

$$\cancel{f_z|_{(4,3,5)} = 1}$$

$$g_x|_{(4,3,5)} = 1$$

$$g_y|_{(4,3,5)} = 2$$

$$g_z|_{(4,3,5)} = 2$$

$$\Delta f = \frac{4}{5}\hat{i} + \frac{3}{5}\hat{j} - \hat{k}$$

$$\Delta g = \hat{i} + 2\hat{j} + 2\hat{k}$$

$\frac{3}{5} \times 2/8/22$

$$\Delta f \cdot \Delta g = \begin{vmatrix} i & j & k \\ 4/5 & 3/5 & -1 \\ 1 & 2 & 2 \end{vmatrix}$$

$$= \hat{i}\left(6/5 + 2\right) - \hat{j}\left(8/5 + 1\right) + \left(8/5 - 3/5\right)$$

$$= \frac{16\hat{i}}{5} - \frac{13\hat{j}}{5} + \frac{5\hat{k}}{5}$$

or

$$\frac{1}{5}\left(16\hat{i} - 13\hat{j} + 5\hat{k}\right)$$ Ans.